

ELECSECURE

Canadian Patent Application Package
(CIPO Draft)

**SMART ELECTRICAL SAFETY AND ENERGY EFFICIENCY
MANAGEMENT SYSTEM WITH IoT-ENABLED ARC FAULT
PROTECTION AND INTEGRATED CLOUD ANALYTICS**

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Jurisdiction	Canada — CIPO
Document version	Draft 1.0 — September 2026
Total claims	20
Technical figures	8

Important: This document is a technical draft for review by a registered Canadian patent agent. It is not legal advice and is not filed with CIPO.

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1. Filing Guide & Checklist

Applicant: Azfar Mushtaq

Invention: Smart Electrical Safety and Energy Efficiency Management System (ElecSecure)

Jurisdiction: Canada — Canadian Intellectual Property Office (CIPO)

Document package version: Draft 1.0 — September 2026

Important legal notice

This package is a **technical draft** prepared to support a Canadian patent application. It is **not legal advice** and does **not** replace consultation with a **registered Canadian patent agent or patent lawyer**. CIPO requires compliance with the **Patent Act** and **Patent Rules** (SOR/2019-251). A professional should review claims, prior art, and formalities before filing.

Documents in this package

Canadian filing steps

1. Prior art search (recommended before filing)

- Search CIPO patent database: <https://www.ic.gc.ca/opic-cipo/cpd/eng/search/basic.html>
- Search Google Patents, Espacenet, and USPTO for: arc fault circuit interrupters, IoT circuit breakers, smart energy management, dual-trip protection devices.
- Document results in a separate prior-art memorandum for your patent agent.

2. Prepare compliant application

- Each section (petition, abstract, description, claims, drawings) must **start on a new page** when filed on paper; for electronic filing via CIPO's portal, upload as separate PDF sections.
- Text: minimum 1.5 line spacing; characters ≥ 0.21 cm height.
- Drawings: black and white, A4, margins (top/left 2.5 cm, right 1.5 cm, bottom 1.0 cm); reference characters ≥ 0.32 cm.
- Language: **English** (or French — do not mix).

3. File with CIPO

Online (recommended): <https://www.ic.gc.ca/opic-cipo/cpd/eng/help/patents/e-filing.html>

Minimum for filing date:

- Petition
- Applicant name and address
- Description
- Claims (at least one)
- Filing fee

Submit soon after filing date:

- Abstract
- Drawings (if invention cannot be understood without them — **recommended for this invention**)
- Inventor name and address
- Statements of entitlement / inventorship

4. Fees (effective January 1, 2026 — verify on CIPO website)

Small entity status: generally Canadian businesses with ≤100 employees and qualifying revenue; confirm eligibility with your agent.

5. Request examination

- Examination must be **requested within 4 years** of the filing date (or PCT national phase entry).
- Without a request for examination, the application will be deemed abandoned.

6. Prosecution timeline (typical)

- First Office Action: often 2–4 years after examination request.
- Respond to objections within **4 months** (extendable).
- Patent term: **20 years** from Canadian filing date (subject to maintenance fees).

Inventorship and ownership

Recommended filing strategy

1. **Canadian national application** — file directly with CIPO using this package.
2. **PCT international application** — if protection in multiple countries is planned within 12 months, file PCT claiming priority from the Canadian filing.
3. **Provisional (US only)** — Canada has no provisional application; early Canadian filing establishes priority.

Post-filing checklist

- [] Record official filing date and application number
- [] Calendar: request examination before 4-year deadline

(Continued in source documents...)

2. Petition for a Patent

TO THE COMMISSIONER OF PATENTS

The undersigned hereby applies for a patent for an invention entitled:

SMART ELECTRICAL SAFETY AND ENERGY EFFICIENCY MANAGEMENT SYSTEM WITH IoT-ENABLED ARC FAULT PROTECTION AND INTEGRATED CLOUD ANALYTICS

Applicant

If applicant is a corporation, substitute corporate name and registered office address.

Inventor(s)

Add co-inventors if applicable. Each inventor's postal address is required by s. 54, Patent Rules.

Statement of Entitlement

The applicant is entitled to apply for and be granted a patent for the invention because:

- ☐ The applicant is the inventor.
- ☐ The applicant is the legal representative of the inventor.
- ☐ The applicant is entitled to apply by virtue of an assignment from the inventor(s).
- ☐ Other: _____

Statement of Inventorship

The person(s) named above as inventor(s) is/are the actual inventor(s) or the legal representative(s) of the actual inventor(s) of the subject-matter defined in the claims.

Priority claim (if applicable)

If filing within 12 months of an earlier application, complete priority details and attach certified copy if requested.

Agent (if represented)

Declaration

I/We declare that:

1. I am/we are the applicant(s) for the patent identified above.
2. The statements made in this petition are true and complete.
3. I/we have reviewed the description, claims, abstract, and drawings submitted with this application.
4. I/we authorize CIPO to correspond with the agent named above (if any).

Signature: _____

Name: Azfar Mushtaq

Date: _____

Capacity: Applicant / Authorized representative

This petition conforms to section 53 of the Patent Rules (SOR/2019-251). Submit as a separate document starting on a new page when filing with CIPO.

3. Abstract

A smart electrical safety and energy efficiency management system comprises an IoT-enabled protective electrical device (ElecSecure), a cloud-based data platform, and a mobile application. The protective device integrates dual tripping mechanisms—a delayed thermal trip for overload conditions and a magnetic trip for short-circuit conditions—with arc fault detection and arc extinguishing via guided arc runners and a multi-plate arc chamber. The device occupies a compact two-module form factor while providing selectable high breaking capacity of 6 kA or 10 kA. Continuous internal self-test, LED fault indication, and bidirectional power connection enhance reliability and serviceability. On-board sensors measure current, voltage, temperature, and arc signatures; a microcontroller processes fault conditions locally and transmits telemetry via Wi-Fi or Bluetooth Low Energy to a cloud platform. The platform stores historical data, generates real-time alerts, and applies machine-learning algorithms to identify energy wastage and deliver personalized efficiency recommendations. A mobile application provides remote monitoring, circuit breaker trip/reset control, data visualization, and integration with smart-home ecosystems. The integrated system improves electrical safety, reduces fire risk from arcing faults, and optimizes energy consumption for residential and commercial installations.

Word count: 149

Abstract must begin on a new page. Must not contain drawings or reference to figure numbers as sole support for features.

4. Description of the Invention

TITLE OF THE INVENTION

SMART ELECTRICAL SAFETY AND ENERGY EFFICIENCY MANAGEMENT SYSTEM WITH IoT-ENABLED ARC FAULT PROTECTION AND INTEGRATED CLOUD ANALYTICS

TECHNICAL FIELD

The present invention relates to electrical protection and energy management systems. More particularly, the invention relates to an integrated system combining an IoT-enabled circuit protection device with arc fault detection and extinguishing, dual tripping mechanisms, continuous self-testing, and cloud-connected analytics for real-time electrical safety monitoring and energy efficiency optimization in residential and commercial installations.

BACKGROUND OF THE INVENTION

Electrical distribution systems in buildings require protection against overload, short-circuit, and arcing fault conditions. Conventional circuit breakers and residual-current devices provide basic overcurrent and ground-fault protection but often lack integrated diagnostics, remote monitoring, and energy analytics capabilities.

Arcing faults—unintentional electrical discharges across an air gap—are a leading cause of residential and commercial electrical fires. Conventional thermal-magnetic breakers may not detect low-level series arcing faults before significant damage occurs. Dedicated arc-fault circuit interrupters (AFCIs) address some arc conditions but are typically deployed as separate devices without integrated energy management or cloud connectivity.

The proliferation of smart home technology and rising energy costs have increased demand for systems that combine electrical safety with real-time energy monitoring, remote control, and data-driven efficiency recommendations. Existing market offerings generally address only subsets of these requirements: standalone smart meters provide consumption data without integrated protection; conventional breakers provide protection without IoT connectivity; and energy management platforms lack embedded arc fault extinguishing hardware in a compact form factor.

There remains a need for a unified system that:

1. Integrates dual thermal-magnetic tripping with active arc fault detection and arc extinguishing in a compact protective device;
2. Provides continuous internal self-testing and localized LED fault indication;
3. Connects securely to a cloud platform for real-time telemetry, alerts, and historical analytics;
4. Delivers machine-learning-based energy efficiency recommendations through a mobile application;
5. Enables remote trip/reset control and smart-home ecosystem integration.

The present invention addresses these needs.

SUMMARY OF THE INVENTION

In a first aspect, the invention provides an IoT-enabled electrical protection device comprising:

- a housing occupying a two-module DIN-rail form factor;
- a delayed thermal tripping mechanism configured to interrupt current flow upon sustained overload;
- a magnetic tripping mechanism configured to interrupt current flow upon short-circuit;
- an arc fault detection subsystem comprising at least one arc sensor and signal processing circuitry;

- an arc extinguishing subsystem comprising arc runners, pre-chamber plates, and an arc chamber configured to divide an electric arc into progressively smaller arc segments until extinguished;
- a microcontroller configured to execute local fault analysis, continuous internal self-test routines, and communication with a connectivity module;
- a connectivity module supporting at least one of Wi-Fi and Bluetooth Low Energy;
- a sensor array comprising current sensors, voltage sensors, and temperature sensors;
- an LED indication subsystem for localized fault identification; and
- a bidirectional power connection interface permitting supply connection from either a top or bottom terminal orientation.

In a second aspect, the invention provides a smart electrical safety and energy efficiency management system comprising the IoT-enabled electrical protection device, a cloud-based data platform, and a mobile application, wherein the cloud platform receives telemetry from one or more devices, stores historical electrical parameter data, generates real-time alerts upon detection of abnormal conditions, and applies machine-learning algorithms to produce personalized energy efficiency recommendations transmitted to the mobile application.

In a third aspect, the invention provides a method of managing electrical safety and energy efficiency comprising: acquiring electrical parameter data from sensors in an IoT-enabled protection device; performing local arc fault detection and tripping when arc signatures exceed predetermined thresholds; transmitting processed telemetry to a cloud platform; analyzing historical and real-time data using machine-learning models; generating alerts and efficiency recommendations; and presenting data and control functions to a user via a mobile application including remote trip and reset commands.

Further aspects and embodiments are defined in the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments of the invention will now be described by way of example with reference to the accompanying drawings, in which:

Figure 1 is a system architecture diagram showing the ElecSecure protection device, connectivity layer, cloud platform, mobile application, and smart-home integrations.

Figure 2 is a front elevational view of the ElecSecure device showing module dimensions, terminals, and LED indicators.

Figure 3 is a sectional side view of the arc extinguishing subsystem showing arc runners (302), pre-chamber plates (304), and arc chamber (306).

Figure 4 is a functional block diagram of internal electronics including microcontroller (402), tripping mechanisms (404, 406), sensors (408), connectivity module (410), and power supply (412).

Figure 5 is a flowchart of the local fault detection and arc extinguishing process.

Figure 6 is a flowchart of cloud data processing, machine-learning analysis, and mobile application interaction.

Figure 7 is a schematic diagram of dual tripping mechanism operation under overload and short-circuit conditions.

Figure 8 is a user interface wireframe of the mobile application showing real-time monitoring, alerts, and remote control panels.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

System Overview (Figure 1)

Reference numeral **100** designates the smart electrical safety and energy efficiency management system. System **100** comprises an ElecSecure protection device **102**, a local network **104**, a cloud platform **106**, a mobile application **108**, and optional smart-home devices **110**.

The protection device **102** is installed in an electrical distribution panel and monitors one or more circuits. Device **102** acquires electrical parameters, performs local protection functions including arc fault detection and extinguishing, and transmits telemetry via connectivity module **112** through local network **104** to cloud platform **106**.

Cloud platform **106** aggregates data from multiple devices **102**, performs storage, analytics, and machine-learning processing, and pushes alerts and recommendations to mobile application **108**. Mobile application **108** provides user interfaces for monitoring, reporting, remote control, and integration with smart-home ecosystems **110**.

ElecSecure Protection Device — Mechanical Design (Figure 2)

The protection device **102** is housed in an enclosure **202** conforming to a two-module DIN-rail form factor, providing a width of approximately 36 mm (two standard 18 mm modules). Despite the compact footprint, the device supports selectable breaking capacities of **6 kA** or **10 kA** in accordance with applicable product standards (e.g., IEC 60898, IEC 61009, or CSA C22.2 No. 5 as adapted for Canadian installations).

Input terminals **204** and output terminals **206** are arranged to permit bidirectional power connection—electrical supply may be connected to either the top or bottom terminal pair, improving installation flexibility in constrained panel layouts.

LED indicators **208** provide localized fault identification. Each LED corresponds to a fault category (e.g., overload, short-circuit, arc fault, ground fault, communication loss, self-test failure), enabling field technicians to diagnose conditions without requiring immediate access to the mobile application.

A manual test/reset interface **210** permits user-initiated self-test cycles and manual reset following a trip event.

Arc Extinguishing Subsystem (Figure 3)

The arc extinguishing subsystem **300** is a distinguishing feature of device **102**. When a fault condition produces an electric arc, arc runners **302** guide the arc from the contact separation point into a pre-chamber defined by pre-chamber plates **304**. The arc is directed into arc chamber **306**, which is configured with insulating partitions and cooling surfaces that subdivide the main arc into a series of smaller arc segments.

Each subdivision increases arc voltage drop and reduces arc current until the arc is extinguished. This active arc management reduces contact erosion, limits thermal damage to adjacent components, and decreases the probability of arc-induced ignition of surrounding materials.

Arc sensors **308** (optical, acoustic, or current-signature-based) detect arc initiation. Upon detection, the microcontroller **402** (Figure 4) commands trip actuation within a predetermined response time (typically less than 100 ms for series arc conditions, subject to applicable standard requirements).

Internal Electronics (Figure 4)

The electronic architecture of device **102** centers on microcontroller **402**, which may be implemented as an ARM Cortex-M class processor or equivalent embedded controller with sufficient processing capacity for real-time sensor sampling, arc signature analysis, and communication protocol handling.

Delayed thermal tripping mechanism 404 comprises a bimetallic element or equivalent thermal actuator calibrated to trip after a sustained overcurrent condition exceeding the rated current by a time-dependent margin (inverse-time characteristic).

Magnetic tripping mechanism 406 comprises a solenoid coil that actuates instantaneously when current exceeds a magnetic trip threshold indicative of a short-circuit condition.

Sensor array **408** includes:

- **Current sensors** (e.g., shunt resistors, current transformers, or Hall-effect sensors) for per-circuit amperage measurement;
- **Voltage sensors** for line voltage and power quality monitoring;
- **Temperature sensors** for thermal monitoring of internal components and ambient panel temperature;
- **Arc sensors** as described above.

Connectivity module **410** supports Wi-Fi (IEEE 802.11 b/g/n) for LAN communication and Bluetooth Low Energy (BLE) for provisioning, local diagnostics, and fallback communication. Firmware in non-volatile memory **414** implements encrypted TLS communication to cloud platform **106**.

Power supply **412** derives low-voltage DC power from the monitored circuit or an auxiliary supply, with voltage regulators maintaining stable operating voltage for electronics independent of line voltage fluctuations within rated range.

(Continued in source documents...)

5. Claims

1. An IoT-enabled electrical protection device comprising:

- a housing configured for mounting on a DIN rail in a two-module form factor;
- a delayed thermal tripping mechanism configured to interrupt electrical current upon detection of a sustained overload condition;
- a magnetic tripping mechanism configured to interrupt electrical current upon detection of a short-circuit condition;
- an arc fault detection subsystem comprising at least one arc sensor and processing circuitry configured to identify an arcing fault based on at least one of current signature, optical emission, or acoustic emission;
- an arc extinguishing subsystem comprising arc runners, pre-chamber plates, and an arc chamber configured to receive an electric arc from the arc runners and to subdivide the electric arc into a plurality of smaller arc segments until the arc is extinguished;
- a microcontroller operatively coupled to the arc fault detection subsystem, the delayed thermal tripping mechanism, and the magnetic tripping mechanism, the microcontroller configured to execute local fault analysis and continuous internal self-test routines;
- a sensor array comprising at least a current sensor and a voltage sensor;
- a connectivity module operatively coupled to the microcontroller and configured to communicate over at least one wireless protocol selected from Wi-Fi and Bluetooth Low Energy; and
- an LED indication subsystem configured to indicate a fault category upon occurrence of a fault condition.

2. The device of claim 1, wherein the housing provides a selectable breaking capacity of at least one of 6 kA or 10 kA.

3. The device of claim 1 or 2, further comprising a bidirectional power connection interface permitting electrical supply connection to either a top terminal pair or a bottom terminal pair.

4. The device of any one of claims 1 to 3, wherein the continuous internal self-test routines comprise at least one of: sensor calibration verification, trip actuator functional verification, communication module heartbeat verification, memory integrity verification, and arc sensor sensitivity verification.

5. The device of any one of claims 1 to 4, wherein the arc fault detection subsystem is configured to command trip actuation within a predetermined response time upon detection of an arcing fault.

6. The device of any one of claims 1 to 5, further comprising a temperature sensor in the sensor array for monitoring at least one of internal component temperature or ambient panel temperature.

7. A smart electrical safety and energy efficiency management system comprising:

- the IoT-enabled electrical protection device of any one of claims 1 to 6;
- a cloud-based data platform configured to receive telemetry from the IoT-enabled electrical protection device, store historical electrical parameter data, and generate alerts upon detection of abnormal electrical conditions; and
- a mobile application configured to communicate with the cloud-based data platform to present real-time electrical parameter data, fault alerts, and remote control commands including at least trip and reset commands for the IoT-enabled electrical protection device.

8. The system of claim 7, wherein the cloud-based data platform comprises a machine-learning module configured to analyze historical and real-time electrical parameter data and to generate personalized energy efficiency recommendations transmitted to the mobile application.

9. The system of claim 7 or 8, wherein the mobile application is configured to integrate with a smart-home ecosystem via at least one of a RESTful API and an MQTT interface to coordinate electrical system functions with at least one external smart-home device.

10. The system of any one of claims 7 to 9, wherein communication between the IoT-enabled electrical protection device, the cloud-based data platform, and the mobile application is encrypted using TLS 1.2 or higher.

11. The system of any one of claims 7 to 10, wherein the mobile application presents data visualization including at least one of graphs, trend reports, and exportable consumption reports derived from telemetry stored on the cloud-based data platform.

12. The system of any one of claims 7 to 11, further comprising earthing and lightning protection components operatively coupled to the cloud-based data platform and configured to report at least one of ground resistance monitoring values and surge protection device status.

13. A method of managing electrical safety and energy efficiency, the method comprising:

- acquiring electrical parameter data from a sensor array of an IoT-enabled electrical protection device installed in an electrical distribution panel;
- performing local fault analysis on the acquired electrical parameter data using a microcontroller of the IoT-enabled electrical protection device;
- detecting an arcing fault based on at least one arc signature derived from the acquired electrical parameter data;
- upon detection of the arcing fault, actuating an arc extinguishing subsystem comprising arc runners, pre-chamber plates, and an arc chamber to extinguish an electric arc and interrupt current flow;
- transmitting processed telemetry from the IoT-enabled electrical protection device to a cloud-based data platform via a wireless connectivity module;
- analyzing at least one of historical and real-time telemetry at the cloud-based data platform using a machine-learning module to generate energy efficiency recommendations;
- generating a fault alert upon detection of an abnormal electrical condition; and
- presenting the energy efficiency recommendations, fault alerts, and remote control functions to a user via a mobile application communicatively coupled to the cloud-based data platform.

14. The method of claim 13, further comprising executing continuous internal self-test routines on the IoT-enabled electrical protection device and reporting self-test results to the cloud-based data platform.

15. The method of claim 13 or 14, further comprising:

- detecting a sustained overload condition and actuating a delayed thermal tripping mechanism; and
- detecting a short-circuit condition and actuating a magnetic tripping mechanism independently of the delayed thermal tripping mechanism.

16. The method of any one of claims 13 to 15, further comprising illuminating an LED indicator on the IoT-enabled electrical protection device to indicate a fault category corresponding to the detected abnormal electrical condition.

17. The method of any one of claims 13 to 16, further comprising receiving user feedback on the energy efficiency recommendations at the cloud-based data platform and updating the machine-learning module based on the user feedback.

18. The method of any one of claims 13 to 17, further comprising integrating the mobile application with a smart-home ecosystem to execute an automation rule triggered by at least one of the fault alert and the energy efficiency recommendations.

19. The device of any one of claims 1 to 6, wherein the at least one wireless protocol comprises Wi-Fi for local area network communication and Bluetooth Low Energy for device provisioning.

20. The system of any one of claims 7 to 12, wherein the cloud-based data platform applies a real-time rule engine to evaluate incoming telemetry against user-configurable thresholds including at least one of voltage sag, voltage swell, phase imbalance, and demand peak conditions.

Total claims: 20 (within standard fee tier; no excess-claim surcharge at filing)

Claims must be clear, concise, and fully supported by the description. Review with a registered Canadian patent agent before filing.

Technical Blueprints & Reference Numerals

The following reference numerals are used consistently across all figures and the patent description.

Numeral	Element
100	Smart electrical safety and energy efficiency management system
102	ElecSecure IoT-enabled protection device
104	Local network (Wi-Fi / LAN)
106	Cloud-based data platform
108	Mobile application
300	Arc extinguishing subsystem
302	Arc runners
304	Pre-chamber plates
306	Arc chamber
402	Microcontroller
404	Delayed thermal tripping mechanism
406	Magnetic tripping mechanism
408	Sensor array
410	Connectivity module (Wi-Fi / BLE)
612	Machine-learning module

FIGURE 1

System Architecture

FIG. 1

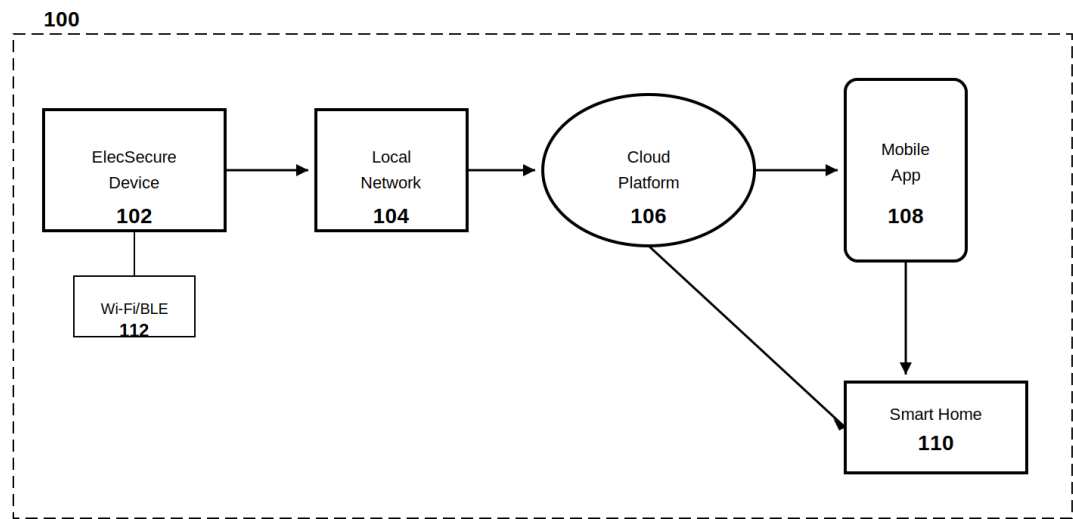


FIG. 1 — End-to-end data flow among protection device (102), local network (104), cloud platform (106), mobile app (108), and smart-home devices (110).

FIGURE 2

Device Front Elevation

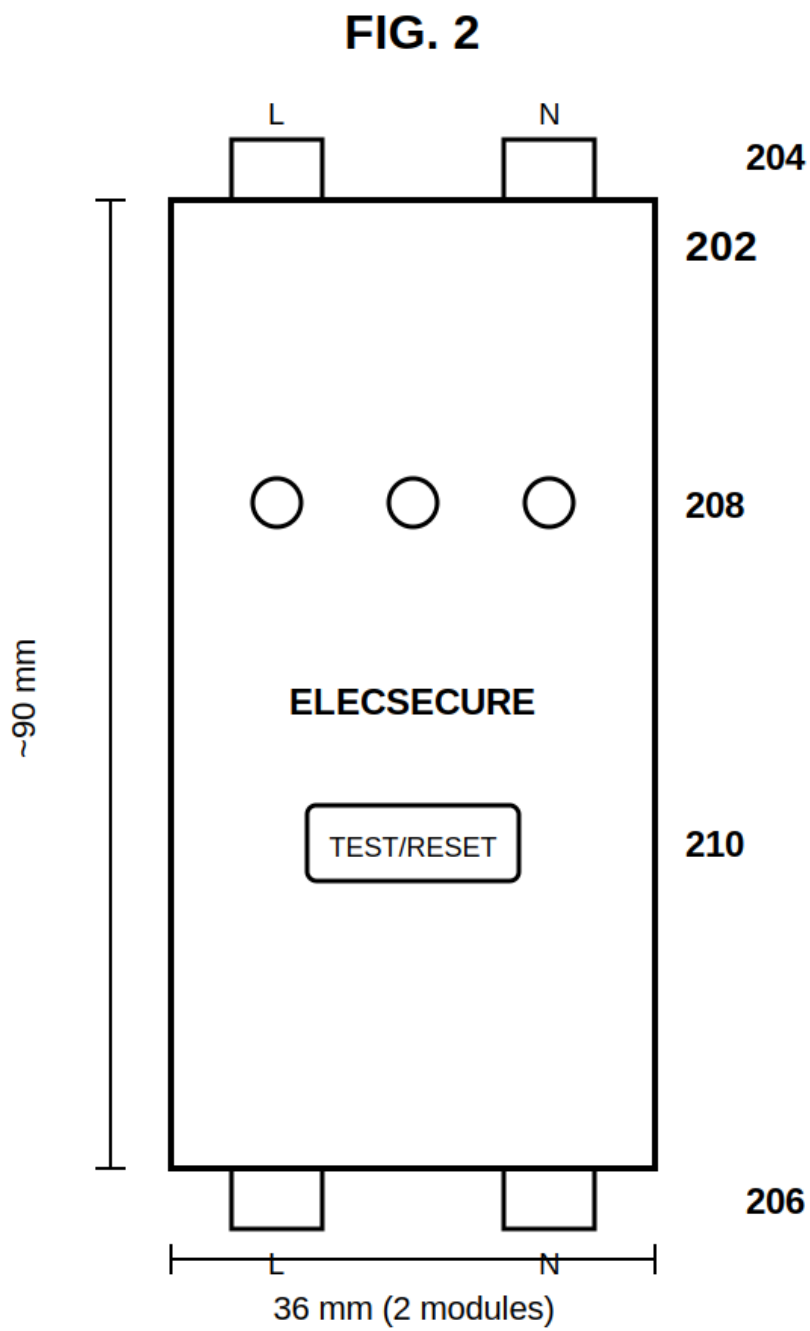


FIG. 2 — Two-module DIN form factor (202) with bidirectional terminals (204, 206), LED fault indicators (208), and test/reset interface (210). Width: 36 mm.

FIGURE 3

Arc Extinguishing Subsystem (Section)

FIG. 3

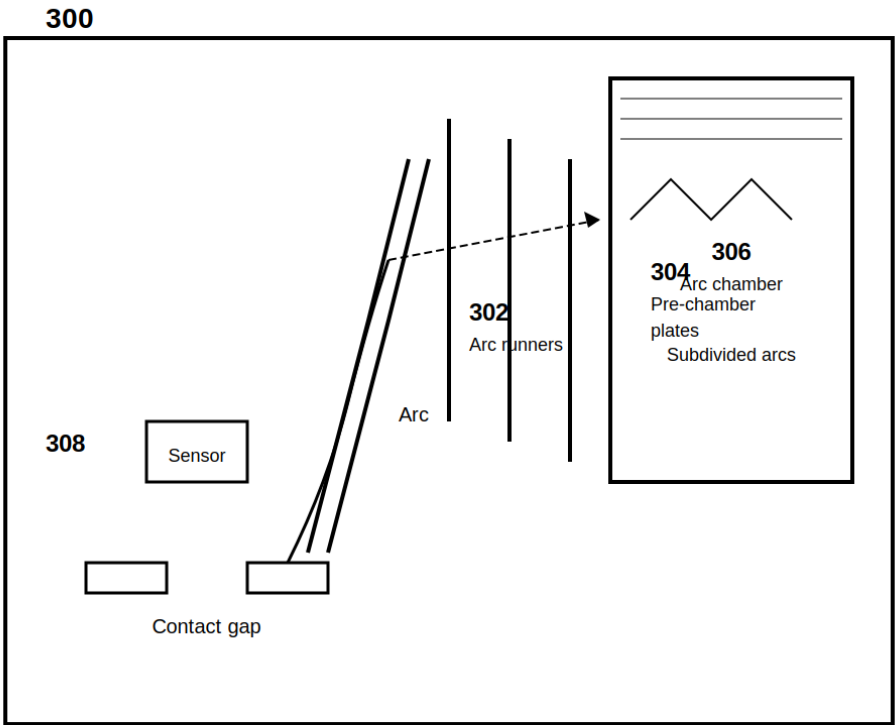


FIG. 3 — Arc runners (302) guide fault arcs through pre-chamber plates (304) into arc chamber (306) for subdivision and extinguishing. Arc sensor (308) shown.

FIGURE 4

Internal Electronics Block Diagram

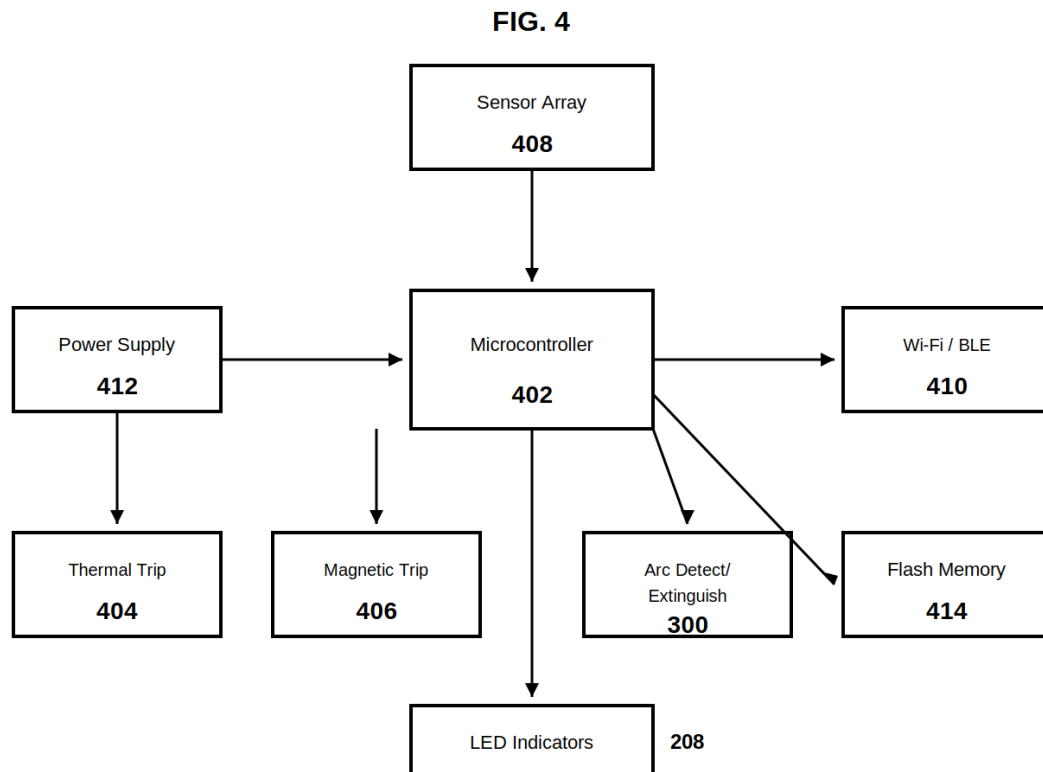


FIG. 4 — Microcontroller (402) interfaces with power supply (412), sensor array (408), trip mechanisms (404, 406), connectivity module (410), memory (414), and arc subsystem (300).

FIGURE 5

Local Fault Detection Flowchart

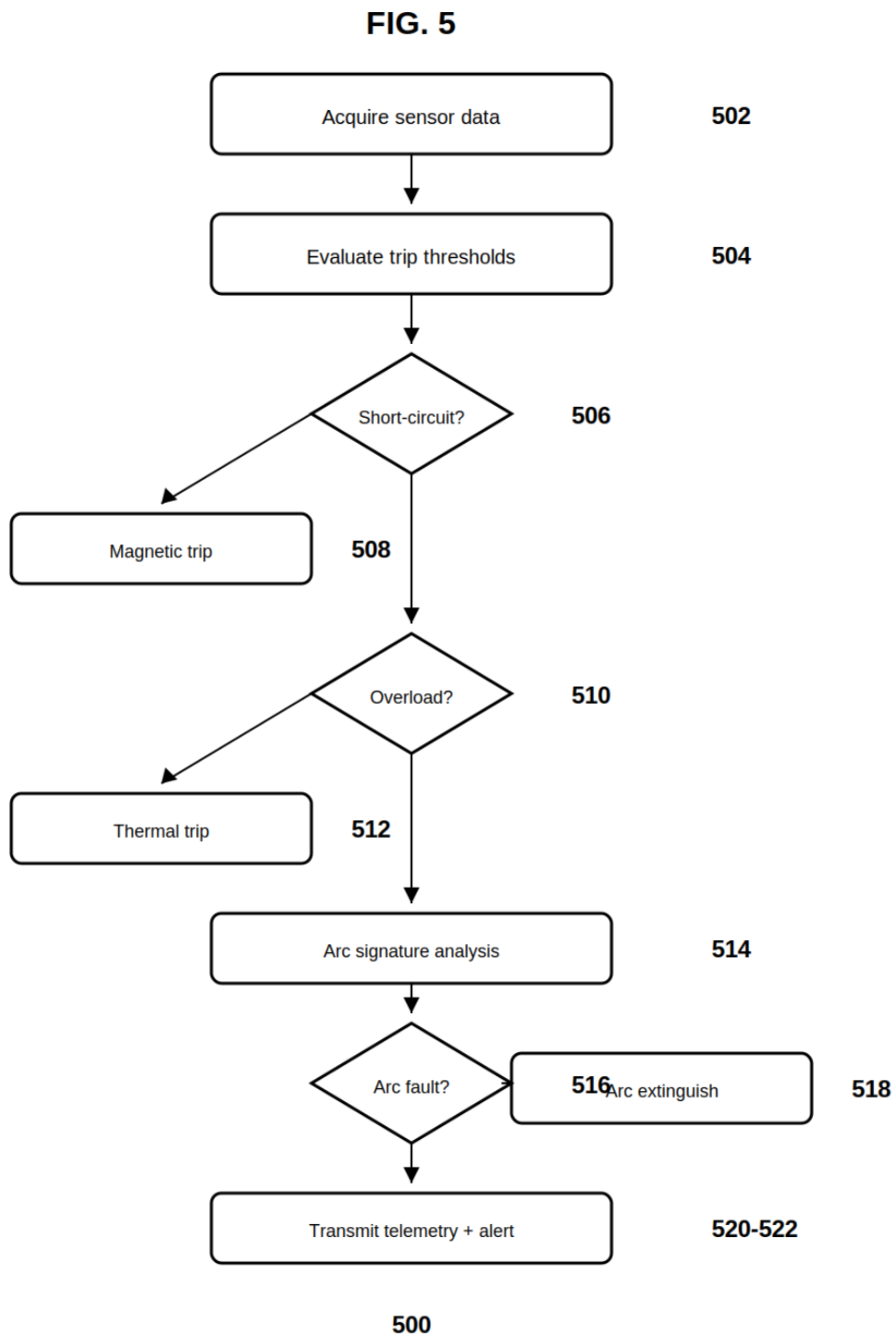


FIG. 5 — Process (500) for sensor acquisition, threshold evaluation, magnetic/thermal tripping, arc analysis, extinguishing, and cloud alert transmission.

FIGURE 6
Cloud Processing Flowchart

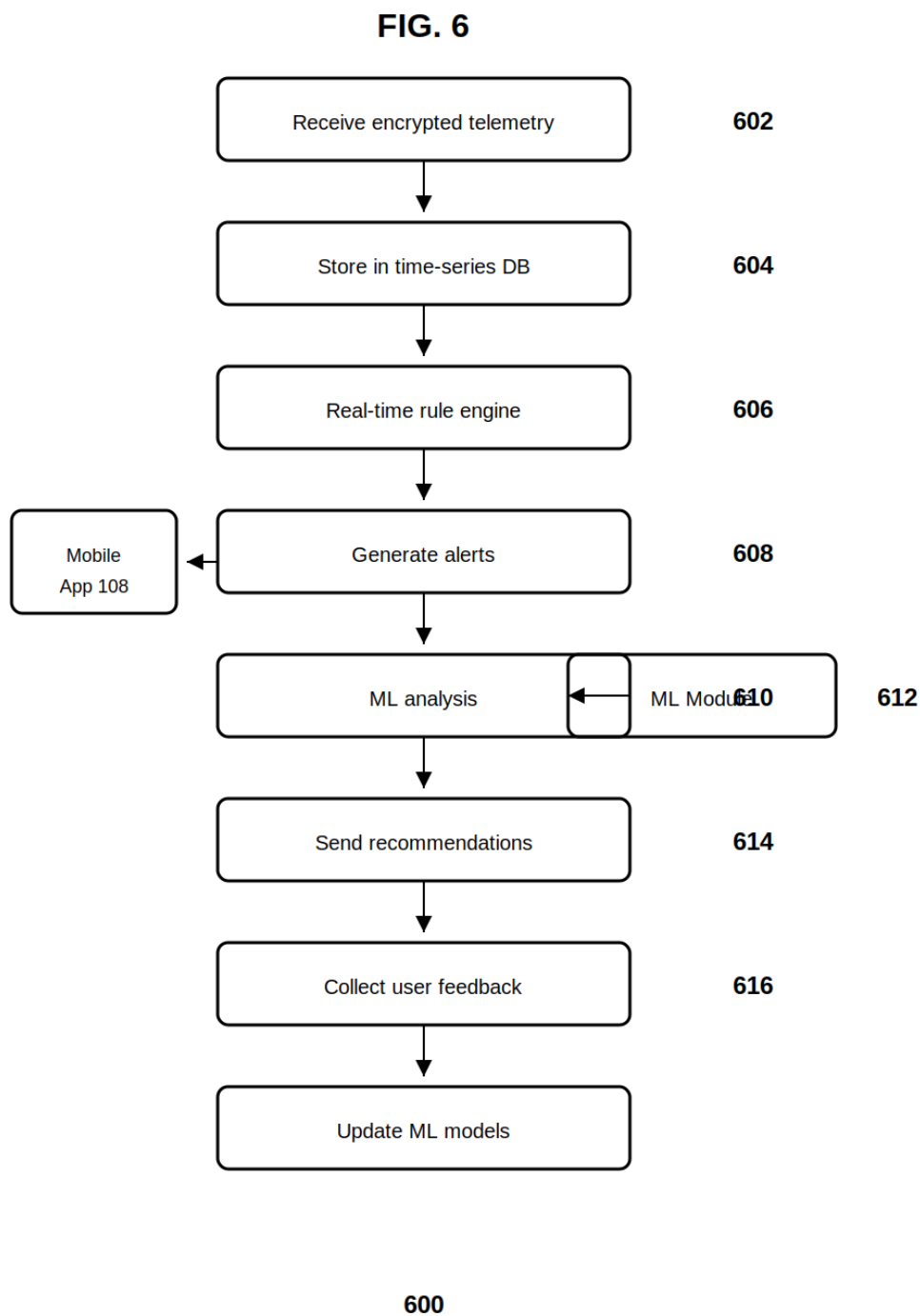
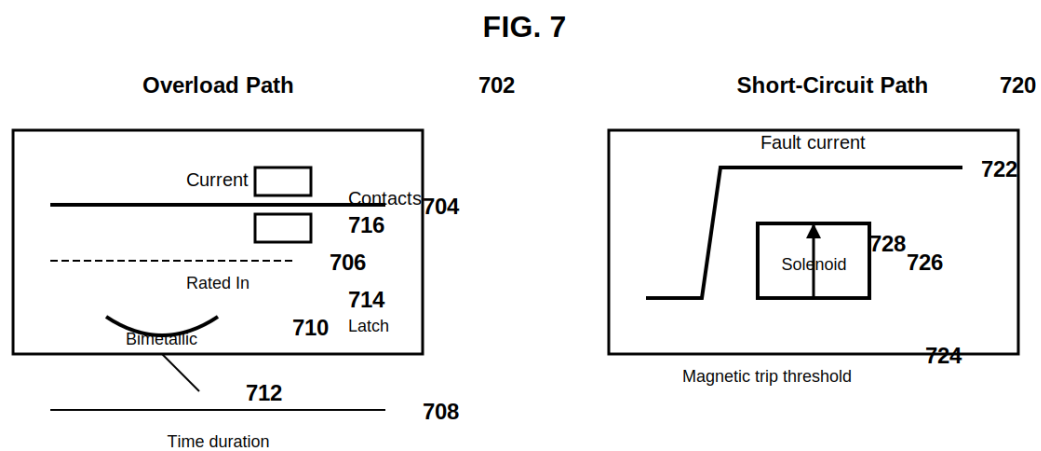


FIG. 6 — Server-side process (600) for telemetry ingestion, storage, rule-based alerts, machine-learning analysis (612), and recommendation delivery.

FIGURE 7
Dual Tripping Mechanism Schematic



Thermal = inverse-time response | Magnetic = instantaneous response

Mechanisms 404 (thermal) and 406 (magnetic) operate independently

FIG. 7 — Independent overload path (702) via bimetallic element (710) and short-circuit path (720) via solenoid (726) actuating contacts (716).

FIGURE 8
Mobile Application Interface

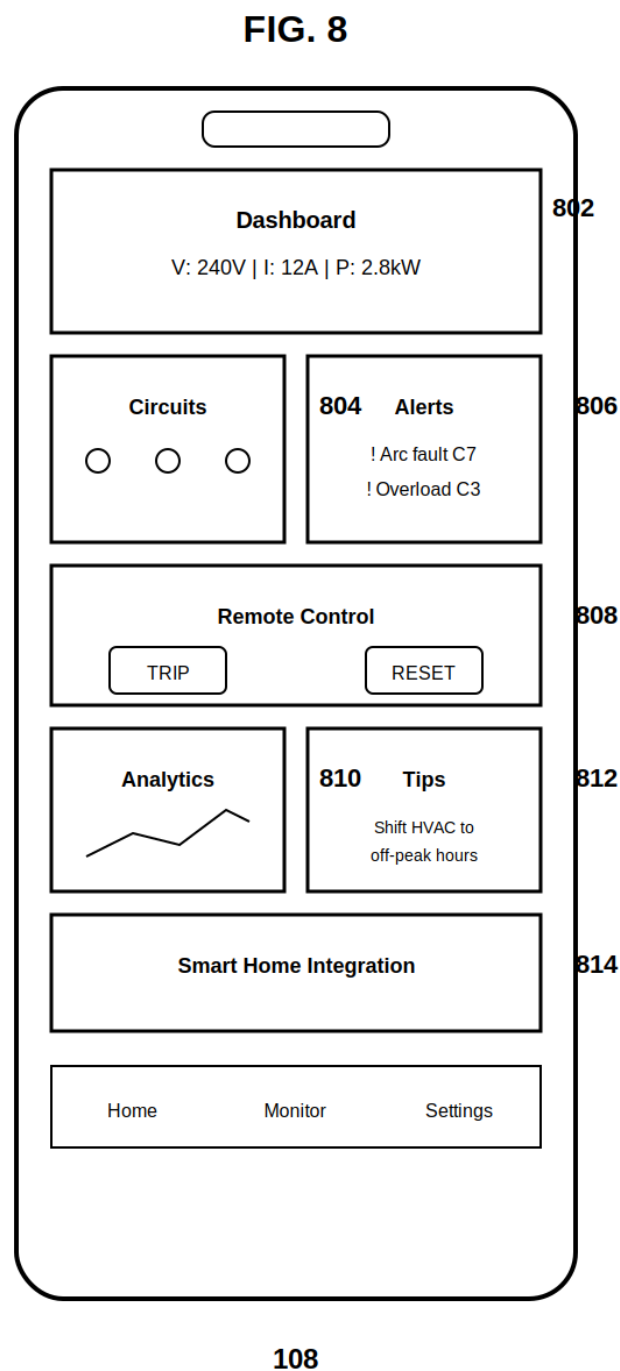


FIG. 8 — User interface (108) with dashboard (802), circuit panel (804), alerts (806), remote control (808), analytics (810), recommendations (812), and smart-home integration (814).

Bill of Materials & Electrical Specifications

Parameter	Value / Range
Rated voltage	120/240 V AC (split-phase, Canadian residential)
Rated current	15 A / 20 A / 30 A (model variants)
Breaking capacity	6 kA or 10 kA @ rated voltage
Thermal trip	Inverse-time, IEC Type B/C curve
Magnetic trip	Instantaneous, 5–10 × I _n
Arc detection response	< 100 ms (target)
Wi-Fi	2.4 GHz, 802.11 b/g/n
BLE	5.0, provisioning service
Operating temperature	–25 °C to +55 °C
Form factor	2-module DIN rail (36 mm width)

Key Components (BOM)

Ref	Component	Qty
U1	Microcontroller (ARM Cortex-M4)	1
U2	Wi-Fi/BLE combo module (ISED-certified)	1
T1	Current transformer / shunt (0–30 A)	1
AS1	Arc sensor (optical / HF current)	1–2
K1	Magnetic trip solenoid	1
BM1	Bimetallic thermal element	1
AR1	Arc runner assembly	1
AC1	Arc chamber housing	1
LED1–4	Status LEDs (fault categories)	4

Canadian Regulatory References

- Canadian Electrical Code (CEC), Part I — CSA C22.1
- CSA C22.2 No. 5 — Molded-case circuit breakers
- CEC Rule 26-720 et seq. — AFCI requirements for dwelling units
- ISED radio certification for Wi-Fi/BLE modules